

## Team Project

### ISyE 6669

#### Part A

##### Question 1

Decision Variables:

- $E_t$  = electricity purchased at hour  $t$  (MWh)
- $H_t$  = hydrogen purchased at hour  $t$  (kg)

Objective Function:

$$\min \sum_{t=1}^{72} (P_t \cdot E_t + 10 \cdot H_t)$$

- $P_t$  = electricity price at hour  $t$  (\$/MWh) (from *Electricity\_Prices.csv*)
- Hydrogen price fixed (\$10/kg)

Constraints:

*Electricity Demand Constraint:*  $E_t + S_t \geq D_t^{elec}$  ,  $\forall t = 1, 2, \dots, N$

- $S_t = 2 \times CF_t$  Solar power available at hour  $t$  (MWh) (solar farm rated at 2MW x Capacity Factor ( $CF_t$ )) (from *Solar Forecast.csv*)
- $D_t^{elec}$  is the electricity demand (from *Electricity\_Demand.csv*)

*Hydrogen Demand Constraint:*  $H_t \geq D_t^{hyd}$  ,  $\forall t = 1, 2, \dots, N$

- $D_t^{hyd}$  is hydrogen demand (from *Hydrogen\_Demand.csv*)

*Non-negativity Constraints:*  $E_t \geq 0$  ,  $H_t \geq 0$  ,  $\forall t = 1, 2, \dots, N$

##### Question 3

**Which constraints are active in the optimal solution?**

- The **electricity demand constraint is active in most cases**. This means the combined electricity purchased ( $E_t$ ) and solar power available ( $S_t$ ) meet the electricity demand ( $D_t^{elec}$ ) exactly for every hour.  
*Evidence:* The electricity gap values calculated during the solution are extremely close to zero (e.g.,  $2.7 \times 10^{-7}$ ), which indicates the constraint is satisfied at its boundary, accounting for numerical precision errors.
- The **hydrogen demand constraint is also active** at all hours. Since hydrogen has unlimited storage capacity, the purchased hydrogen ( $H_t$ ) always equals the hydrogen demand ( $D_t^{hyd}$ ).

*Evidence:* The hydrogen gap values are also near zero (e.g.,  $-1.0 \times 10^{-9}$ ), indicating the demand is met exactly.

- **Solar Utilization:**  
Solar power ( $S_t$ ) is always fully utilized when available, as it is a zero-cost resource.

#### **Would the solution differ for a shorter time horizon?**

Yes, a shorter time horizon (e.g., 24 or 48 hours) would limit the model's ability to optimize costs over time. The current 72-hour model can strategically purchase electricity during cheaper hours later in the time frame, while a shorter horizon forces immediate purchases, potentially at higher costs. Hydrogen purchases are less affected due to unlimited storage, but electricity, without storage, would result in more frequent active constraints and higher immediate costs.

#### **Task Allocation**

- **Question 1:** Mahsa and Vedika individually contributed to formulating the problem and defining the variables and constraints.
- **Question 2:** Vedika developed the main Python code to solve the optimization problem. Parisa and Mahsa verified the results.
- **Question 3:** Mahsa analyzed the solution, interpreted the results, and provided a discussion on active constraints and time horizon impacts.